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INSY6112 ASSIGNMENT 1

Group: 1

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Question 1

When choosing a database for a system it is important to consider many factors, such as the structure of the data, the types of data, the scalability of the system and the volume of data that will be encountered. I believe the best type of database for a South African metropolitan municipality would be a noSql database. Also known as a not only sql database.

NoSql databases are type of database that uses either unstructured or semi structured data (Coronel et al., 2020). Unlike a traditional relation database, which has tightly structured data that uses rows and columns in the form of tables (Coronel et al., 2020) the noSql may instead store the data in the form of value key pairs (Coronel et al., 2020). A benefit of this is that it allows the storage of data that cannot normally stored in a relational database such as photos, emails, xml and Json objects (Coronel et al., 2020). These sorts of databases are also a lot easier to use as they might not need the user know a query language such as sql.

One of the reasons I suggest noSql databases for a city municipality is that noSql databases are best suited when a large volume of data is stored (Coronel et al., 2020) and since the database would need to capture readings for every household roughly every 30 minutes a relational database would not be able to handle data a such high volumes. NoSql databases are also favoured when scalability is a factor (Coronel et al., 2020). As the system will need to be able to keep up with a growing population this is extremely important. Relational databases are usually favoured when a finite number of users and applications need access to the database (Coronel et al., 2020) however a database for a municipality needs to be accessed by an unknown number of users, departments and applications such as accounts, attorneys, auditors and varied billing / reporting software throughout the city, as such a noSql database is better than a relational database in this regard.

When talking about big data it is normally broken down into three categories, volume, velocity and variety (Coronel et al., 2020). Volume is how much data is recorded (Coronel et al., 2020). In this case every household in the city needs records kept. Velocity is the rate the data is stored (Coronel et al., 2020). So smart readers sending data every 30 minutes is the rate and lastly variety is how many different types of data is needed (Coronel et al., 2020). Data for electrical systems, water systems, sanitation services, houses, payments, revenue reports are some of the different types of data need to be stored.

Four different types of noSql databases include: Key value databases, document databases, column- oriented databases and graph databases (Coronel et al., 2020). Key value databases

can store almost any type of content such as xml files and images. The content (the value) can be accessed by using its unique identifier (the key (Amazon, 2023). There are no foreign keys in a value key database, so the data store is not linked together (Coronel et al., 2020). The data is stored is what is known as bucket (Coronel et al., 2020). Some examples of key value database include Dynamo, Riak and Redis (Coronel et al., 2020).

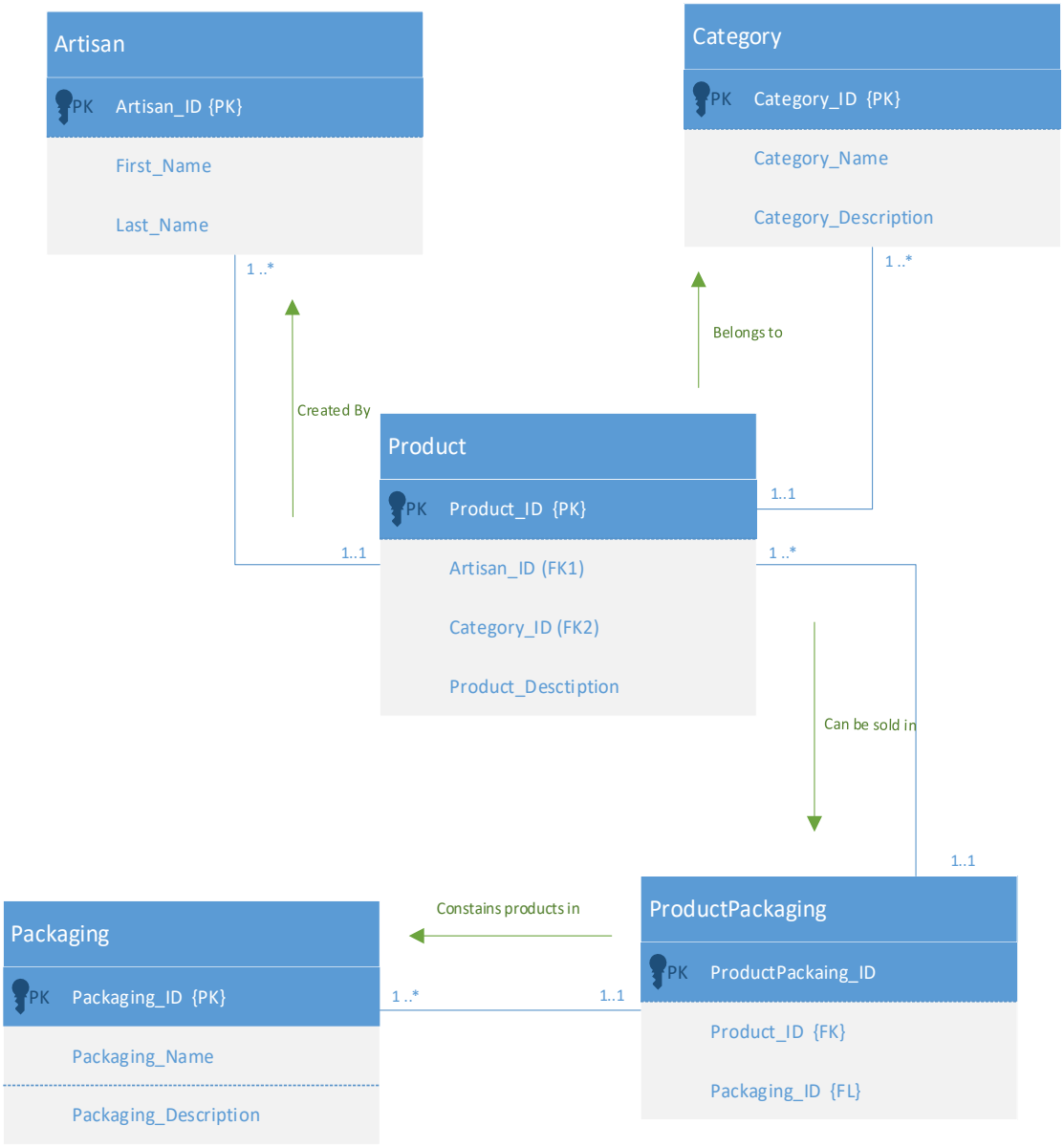
Document databases are like key value databases except they can only store documents. Some examples of documents that can be stored are JSON files and xml files (Coronel et al., 2020). Another difference is that records in a document database can be linked and queried with the use of tags (Coronel et al., 2020). Data in a document database is stored in what is called collections (Coronel et al., 2020). Some examples include: MongoDb, RavenDb and CouchDb (Coronel et al., 2020).

Graph databases are a type of database that uses graph theory. It uses mathematic concepts to model and store the data (Coronel et al., 2020). These types of databases are often used when the data has complex relationships that needs to be tracked. In such cases the relationships might be considered just as important as the data they link (Coronel et al., 2020). A common use case for graph databases is for social networking websites. An example of a graph database is AranoDb and GraphBase (Coronel et al., 2020).

The last type of noSql database is the column-oriented database. These are like the tradition relational databases except they use orientate their data around the columns instead of the rows (Coronel et al., 2020). An advantage of this type of database is that it is more efficient and less resource intensive to query data (Putra, 2025). This is because when data is fetched only the required columns are fetched as opposed to row-orientated databases that will fetch the entire row, even if some columns are not needed. (Coronel et al., 2020)

Since a city might need to store such high amounts of data and at such a high frequency as well as need to store many different types of data but at the same time need to be able to link and query relating records a document database stored Json files might be the best type of database that a large and growing city can use.

Question 2


Artisan
PK Artisan_ID {PK}
First_Name
Last_Name
Category
PK Category_ID {PK}
Category_Name
Category_Description
1 ..*
Created By
Belongs to
1..*
Product
PK Product_ID {PK}
Artisan_ID (FK1)
Category_ID (FK2)
Product_Desctiption
1..1
1 ..*
Can be sold in
1..1
Packaging
PK Packaging_ID {PK}
Packaging_Name
Packaging_Description
Constains products in
1 ..*
1..1
ProductPackaging
PK ProductPackaing_ID
Product_ID {FK}
Packaging_ID {FL}

Question 3

Q3.1

Five possible entities for the fitness centre can include:

1. Member
2. Class
3. Instructor
4. Booking
5. Payment_History

Q3.2

The member table can include the following attributes: Member Id, Active, First name and Last name

The Class entity can have the following attributes: Class Id, name, Instructor Id

The Instructor entity can have the following attributes: Instructor Id, First name, Last name

The Booking entity can have the following: Booking Id, Member Id, Class Id, Time and Date of Class

The payment history entity can have the following: Payment History Id, Member Id, Fee Total and Payment amount

Q3.3

The Member and the Booking entities have a one to many relationship with each member being able to have multiple bookings. However, each booking will have only one member linked to it, as well as having one class linked.

References

Amazon, 2023. [online] Relational vs nonrelational databases - difference between types of databases - AWS. Available at: <https://aws.amazon.com/compare/the-difference-between-relational-and-non-relational-databases/> [Accessed 22 Aug. 2026].

Coronel, C., Blewitt, C., Crockett, K. and Morris, S., 2020. *Database principles: Fundamentals of design, implementation, and management*. 3rd ed. Australia: Cengage.

Putra, A., 2025. *What is a Columnar Database? (vs. Row-oriented Database)* . [online] YouTube. Available at: <https://www.youtube.com/watch?v=1MnvuNg33pA> [Accessed 22 Aug. 2026].